Huawei Access Node QoS/Congestion Event Troubleshooting and Resolution Procedure for MAR-ACC-26

Node: MAR-ACC-26 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_3623

Description

The MAR-ACC-26 node, situated at the Marathahalli Core site, operates as an access role within the network architecture. Given its function, this node is susceptible to Quality of Service (QoS) and congestion events, such as microbursts, interface tail drops, and QoS queue starvation. These issues can arise due to a combination of factors including, but not limited to, improper QoS policy configurations, insufficient bandwidth allocation, and hardware limitations. The Huawei equipment, with its proprietary CLI dialect, necessitates a tailored approach for diagnostic and resolution commands. The ALARM_3623, classified as a minor severity event, indicates a need for prompt investigation to prevent escalation into more severe network disruptions. Understanding the underlying protocol mechanics, such as the queuing algorithms and scheduling mechanisms implemented in the Huawei devices, is crucial for effective troubleshooting. The interaction between these mechanisms and the configured QoS policies can lead to congestion events if not properly aligned with the network's traffic patterns and requirements. Thus, a comprehensive troubleshooting procedure must consider both the hardware capabilities and the software configurations of the device.

Pre-requisite Checks

- Verify the physical integrity of the node and its connections at the Marathahalli Core site.
- Check the status of redundant links to ensure no single point of failure exists.
- Confirm that all necessary diagnostic tools and access credentials are available.

Potential Root Causes

Cause	Probability
Improper QoS Policy Configuration	High
Insufficient Bandwidth Allocation	Medium
Hardware Limitations or Failures	Low

Diagnostic Commands & Expected Outputs

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Command: display qos policy, Expected Output: List of applied QoS policies and their parameters., Description: This command isolates the current QoS policy configurations to identify any misalignments with the network's traffic requirements or improper settings that could lead to congestion events.
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Command: display interface brief, Expected Output: Summary of interface statuses, including any errors or drops., Description: This command helps in identifying interfaces experiencing errors, drops, or other issues that could be contributing to the congestion event.

Command: display traffic-statistics interface, Expected Output: Detailed traffic statistics for each interface., Description: This command provides insights into the traffic patterns and volumes, aiding in the identification of potential bottlenecks or misconfigurations leading to congestion.
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Resolution Steps

Step 1: Step: Adjust QoS Policy Configuration, Command: qos policy classifier behavior , Description: Modify the QoS policy to ensure it aligns with the network's traffic requirements, adjusting parameters such as queue scheduling, policing, and shaping as necessary to prevent congestion events.

Step 2: Step: Allocate Sufficient Bandwidth, Command: interface bandwidth , Description: Increase the bandwidth allocation for the affected interface to accommodate the required traffic volume, preventing congestion due to insufficient bandwidth.

Step 3: Step: Upgrade or Replace Hardware, Command: Not applicable via CLI; requires physical intervention., Description: If hardware limitations or failures are identified as the root cause, upgrading or replacing the affected hardware components may be necessary to resolve the issue.

Post-Check Verification

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Command: display qos policy, Expected Output: Updated list of applied QoS policies reflecting changes made., Description: Verify that the QoS policy configurations have been successfully updated to reflect the adjustments made during the resolution steps.

Command: display interface brief, Expected Output: Interface statuses showing no errors or drops., Description: Confirm that the interfaces are no longer experiencing errors or drops, indicating a resolution to the congestion event.

Command: display traffic-statistics interface, Expected Output: Traffic statistics showing improved traffic flow and no congestion., Description: Verify that the traffic patterns and volumes have normalized, with no signs of congestion or bottlenecks.
```

Roll-back Procedures

- Step: Revert QoS Policy Configuration, Command: qos policy classifier behavior , Description: If the adjustments made to the QoS policy configurations resulted in undesirable outcomes, revert to the original policy settings to restore the previous state.

- Step: Revert Bandwidth Allocation, Command: interface undo bandwidth, Description: If the bandwidth allocation changes did not resolve the issue or introduced new problems, revert the bandwidth settings to their original values.

- Step: Revert Hardware Changes, Command: Not applicable via CLI; requires physical intervention., Description: If hardware upgrades or replacements were made, and these changes did not resolve the issue or introduced new problems, revert to the original hardware configuration, which may require physical intervention.